

Daolang Huang

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Summary

I am currently a final-year PhD student. My research primarily focuses on **amortized inference** – developing advanced neural networks (e.g., neural processes, prior-data fitted networks) to accelerate *Bayesian inference* and *sequential decision-making* tasks, with a strong interest in Bayesian experimental design, Bayesian optimization, simulation-based inference, tabular foundation models, and reinforcement learning.

Education

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|---|------------------------------|
| Aalto University
<i>PhD, Doctoral programme in Science.</i> <ul style="list-style-type: none"> Supervisors: Prof. Samuel Kaski (Aalto University & University of Manchester), Prof. Luigi Acerbi (University of Helsinki) Fully funded by Finnish Center for Artificial Intelligence (FCAI), affiliated with ELLIS Institute Finland. | Jan 2023 – present |
| University of Oxford
<i>Visiting PhD Student</i> <ul style="list-style-type: none"> Conducted research on Bayesian experimental design and tabular foundation models within the RainML Group, under the supervision of Prof. Tom Rainforth | Feb 2026 – May 2026 |
| Aalto University
<i>MSc, Machine Learning, Data Science and Artificial Intelligence.</i> <ul style="list-style-type: none"> GPA: 4.68/5 (Major 5/5), graduated with honours Master's thesis: Bayesian Optimization Augmented with Actively Elicited Expert Knowledge | Sept 2020 – July 2022 |
| Jinan University
<i>BSc, Computer Science and Technology.</i> <ul style="list-style-type: none"> GPA: 87/100 (Major 89/100) Bachelor thesis: Multi-Modal Fusion for Fake News Detection | Sept 2016 – July 2020 |

Experience

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|---|--|
| Research Scientist Intern
<i>Meta</i> <ul style="list-style-type: none"> Working in the Monetization team, conducting research related to ads ranking and experimental design. | June 2026 – Present
<i>New York, US</i> |
| Doctoral Researcher
<i>Aalto University</i> <ul style="list-style-type: none"> Doctoral researcher in the Probabilistic Machine Learning group (Aalto University) and the Machine and Human Intelligence group (University of Helsinki), affiliated with ELLIS Institute Finland and FCAI Topic: Intelligent Bayesian inference and data acquisition via amortization | Jan 2023 – Present
<i>Espoo, Finland</i> |
| Research Assistant
<i>Aalto University</i> <ul style="list-style-type: none"> Research assistant in the Probabilistic Machine Learning group, supervised by Prof. Samuel Kaski Worked on Knowledge elicitation and human-in-the-loop Bayesian optimization | Oct 2021 – Dec 2022
<i>Espoo, Finland</i> |
| Machine Learning Research Intern
<i>Aalto University</i> <ul style="list-style-type: none"> Summer intern in the Probabilistic Machine Learning group, supervised by Prof. Samuel Kaski Worked on Targeted Bayesian active learning | June 2021 – Aug 2021
<i>Espoo, Finland</i> |

Founder & Manager

EDM Heaven

- Specialized in customized music services for musicians and producers, including mixing and mastering, catering to diverse client requirements and genres

Feb 2017 – Aug 2020

Guangzhou, China

Software Engineering Intern

Neusoft

- Summer intern, Python developer

June 2019 – Aug 2019

Guangzhou, China

Publications

First-author and co-first-author conference papers

1. **Daolang Huang**, Xinyi Wen, Ayush Bharti, Samuel Kaski*, Luigi Acerbi*. “ALINE: Joint Amortization for Bayesian Inference and Active Data Acquisition”. In *Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS)*, 2025. **Spotlight (Top 3.2%)**
2. Xinyu Zhang*, **Daolang Huang***, Samuel Kaski, Julien Martinelli. “PABBO: Preferential Amortized Black-Box Optimization”. In *The 13th International Conference on Learning Representations (ICLR)*, 2025. **Spotlight (Top 5.1%)**
3. Paul Chang*, Nasrulloh Loka*, **Daolang Huang***, Ulpur Remes, Samuel Kaski, Luigi Acerbi. “Amortized Probabilistic Conditioning for Optimization, Simulation and Inference”. In *The 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
4. **Daolang Huang**, Yujia Guo, Luigi Acerbi, Samuel Kaski. “Amortized Bayesian Experimental Design for Decision-Making”. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
5. **Daolang Huang***, Ayush Bharti*, Amauri Souza, Luigi Acerbi, Samuel Kaski, “Learning Robust Statistics for Simulation-based Inference under Model Misspecification”. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
6. **Daolang Huang**, Manuel Haussmann, Ulpur Remes, ST John, Grégoire Clarté, Kevin Sebastian Luck, Samuel Kaski, Luigi Acerbi. “Practical Equivariances via Relational Conditional Neural Processes”. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NeurIPS)*, 2023.

Other conference papers

1. Yujia Guo, **Daolang Huang**, Xinyu Zhang, Sammie Katt, Samuel Kaski, Ayush Bharti, “Constrained Bayesian Experimental Design via Online Planning”. In *Forty-third International Conference on Machine Learning (ICML)*, 2026.
2. Xinyu Zhang, Conor Hassan, Julien Martinelli, **Daolang Huang**, Samuel Kaski, “In-Context Multi-Objective Optimization”. In *The 14th International Conference on Learning Representations (ICLR)*, 2026.
3. Yang Yang, Severi Rissanen, Paul E. Chang, Nasrulloh Loka, **Daolang Huang**, Arno Solin, Markus Heinonen, Luigi Acerbi. “PriorGuide: Test-Time Prior Adaptation for Simulation-based Inference”. In *The 14th International Conference on Learning Representations (ICLR)*, 2026.
4. Conor Hassan, Nasrulloh Loka, Cen-You Li, **Daolang Huang**, Paul E. Chang, Yang Yang, Francesco Silvestrin, Samuel Kaski, Luigi Acerbi. “Efficient Autoregressive Inference for Transformer Probabilistic Models”. In *The 14th International Conference on Learning Representations (ICLR)*, 2026.
5. Ayush Bharti, **Daolang Huang**, Samuel Kaski, F.X Briol, “Cost-Aware Simulation-Based Inference”. In *The 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
6. Runkai Zheng, Yinqi Zhang, **Daolang Huang**, Qingliang Chen. “Sequential Convolution and Runge-Kutta Residual Architecture for Image Compressed Sensing”. In *European Conference on Computer Vision (ECCV)*, 2020.

Workshop papers and preprints

1. Yujia Guo, **Daolang Huang**, Xinyu Zhang, Sammie Katt, Samuel Kaski, Ayush Bharti, “Constraint-aware Bayesian Experimental Design”. In *ICML 2026 Workshop: Epistemic Intelligence in Machine Learning (ICML-EIML)*, 2026.
2. Conor Hassan, Nasrullo Loka, Cen-You Li, **Daolang Huang**, Paul E. Chang, Yang Yang, Francesco Silvestrin, Samuel Kaski, Luigi Acerbi. “Efficient Autoregressive Inference for Tabular Foundation Models”. In *EurIPS 2025 Workshop AI for Tabular Data*, 2025.
3. Paul E. Chang, Severi Rissanen, Nasrullo Loka, **Daolang Huang**, Luigi Acerbi. “Inference-Time Prior Adaptation in Simulation-Based Inference via Guided Diffusion Models”. In *ICLR Workshop 2025 on Frontiers in Probabilistic Inference: Sampling Meets Learning (ICLR-FPI)*, 2025.
4. **Daolang Huang**, Yujia Guo, Luigi Acerbi, Samuel Kaski. “Amortized Decision-Aware Bayesian Experimental Design”. In *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty (NeurIPS-BDU)*, 2024.
5. **Daolang Huang**, Louis Filstroff, Petrus Mikkola, Runkai Zheng, Milica Todorovic and Samuel Kaski. “Augmenting Bayesian Optimization with Preference-based Expert Feedback”. In *ICML 2023 Workshop The Many Facets of Preference-Based Learning (ICML-MFPL)*, 2023.
6. **Daolang Huang**, Louis Filstroff, Petrus Mikkola, Runkai Zheng, Samuel Kaski. “Bayesian Optimization Augmented with Actively Elicited Expert Knowledge”. *arXiv:2208.08742*, 2022.
7. **Daolang Huang**, Ricardo Falcon Perez. “Sseldnet: a fully end-to-end sample-level framework for sound event localization and detection”. In *Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2021.

*Equal contribution

Teaching

Master’s Thesis Supervisor, Zhanghu Zhao, *Aalto University*, Autumn 2025.

Research Assistant Supervisor, Ruifeng Nie, *University of Helsinki*, Spring 2025.

Summer Intern Supervisor, Xinyi Wen, *Aalto University*, Summer 2024.

Master’s Thesis Supervisor, Xinyu Zhang, *Aalto University*, Spring 2024.

Bachelor’s Thesis Supervisor, Filip Eller, *Aalto University*, Spring 2024.

Bachelor’s Thesis Supervisor, Petrus Nikoskinen, *Aalto University*, Spring 2024.

Teaching Assistant, Deep learning, *Aalto University*, Spring 2024.

Tutor, Seminar in computer science, *Aalto University*, Autumn 2023.

Teaching Assistant, Deep learning, *Aalto University*, Spring 2023.

Teaching Assistant, Deep learning, *Aalto University*, Spring 2022.

Teaching Assistant, C Programming, *Jinan University-University of Birmingham Joint Institute*, Spring 2020.

Events

Invited Talk, *RainML Lab, University of Oxford*, October 2025.

Invited Talk, *The Seminar on Computational Engineering, LUT University*, September 2025.

Invited Talk, *Accelerating statistical inference and experimental design with machine learning workshop, Isaac Newton Institute, Cambridge*, June 2025.

Invited Talk, *Mathematical perspective on machine learning seminar, University of Helsinki*, November 2023.

Talk, *FCAI AI day*, November 2023.

Talk and Poster, *ELLIS Doctoral Symposium 2023*, August 2023.

Academic Services

NeurIPS 2024, 2025, 2026; AISTATS 2025, 2026; ICLR 2025, 2026; ICML 2025, 2026 (**Gold Reviewer**); TMLR

Scholarships and Awards

The Encouragement Grant, The Finnish Foundation for Technology Promotion, 2025.

Incentive Scholarships, Aalto University, 2025.

Top 3 Nominee for AI Researcher of the Year, AI Finland, 2024.

Full PhD Funding, Finnish Center for Artificial Intelligence, 2023.

Dean's List Scholarships, Aalto university, 2021.

International Talent Program, Aalto University, 2021.

Honours Program in Computer Science, Aalto University, 2022.

Aalto University Scholarship (100% tuition waiver), Aalto University, 2020.

Scholarship for Outstanding Students, Jinan University, 2017, 2018, 2019.